

DATA 581

Modeling and Simulation II



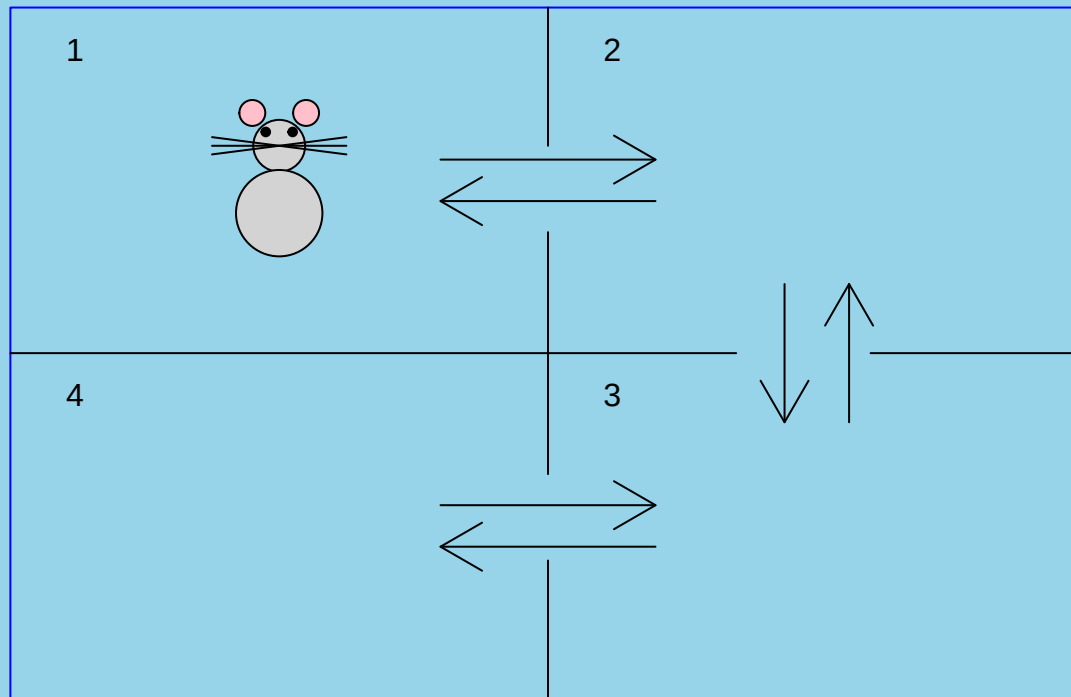
Discrete Time Markov Chains

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Example

A Mouse in a Maze



The State Variable

We consider the times when the mouse moves from one compartment to another, denoting these as $t = 0, 1, 2, 3, \dots$

We let X_t denote the location of the mouse at time t , the compartment or *state* of the process.

If the mouse starts in compartment 1, and enters compartment 2, then compartment 3, back to 2, back to 3, then 4, and back to 3.

$$X_0 = 1, X_1 = 2, X_2 = 3, X_3 = 2, X_4 = 3, X_5 = 4, X_6 = 3, \dots$$

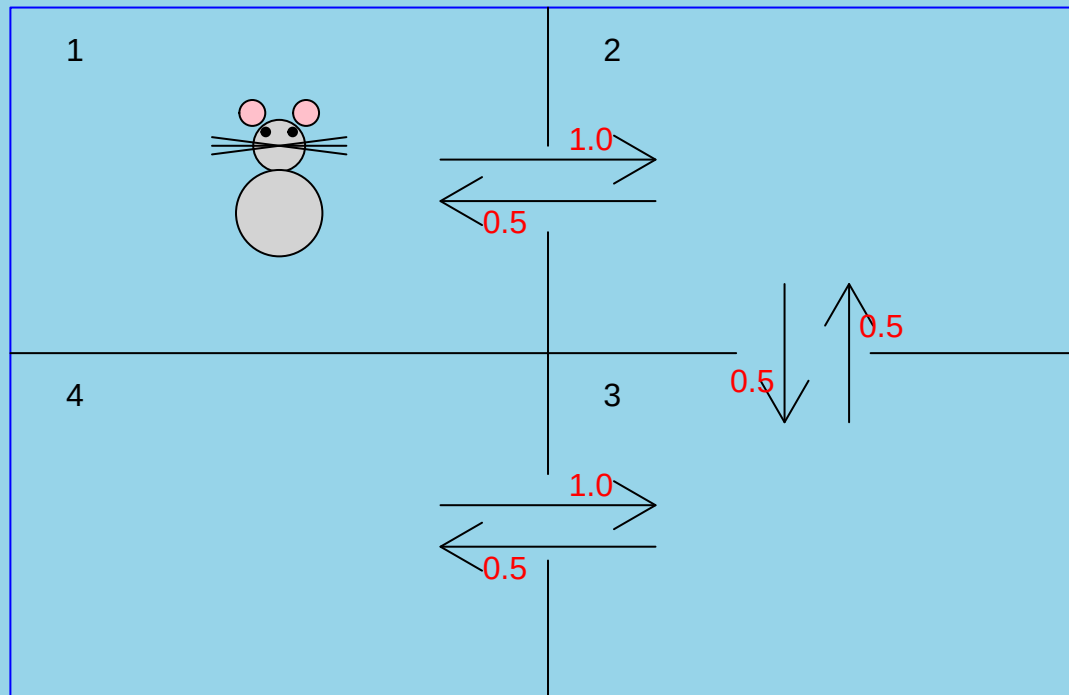
A Mouse Movement Model - Complete Randomness

If the mouse wanders aimlessly, we might attach probabilities to the *transitions* between compartments in the maze as in the following diagram.

Alternatively, the mouse might favour one of the compartments over another, perhaps because of the location of food or a particular odor.

A Mouse Movement Model - Complete Randomness

Transition Probabilities



A Mouse Movement Model - Complete Randomness

Transition Matrix

The transition probabilities can be organized systematically into a matrix, noting that impossible transitions have probability 0.

The 1st row of the matrix lists the transition probabilities from the first compartment to all other compartments. We assume that the mouse is not staying in compartment 1: 0, 1, 0, 0.

2nd row – transition probabilities out of compartment 2: 0.5, 0, 0.5, 0.

3rd row – transition probabilities from compartment 3: 0, 0.5, 0, 0.5.

4th row – transition probabilities out of compartment 4: 0, 0, 1, 0.

A Mouse Movement Model - Complete Randomness

Transition Matrix

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0.5 & 0 & 0.5 & 0 \\ 0 & 0.5 & 0 & 0.5 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Entry (i, j) represents the probability of transition into compartment j , given that the mouse was in state i .

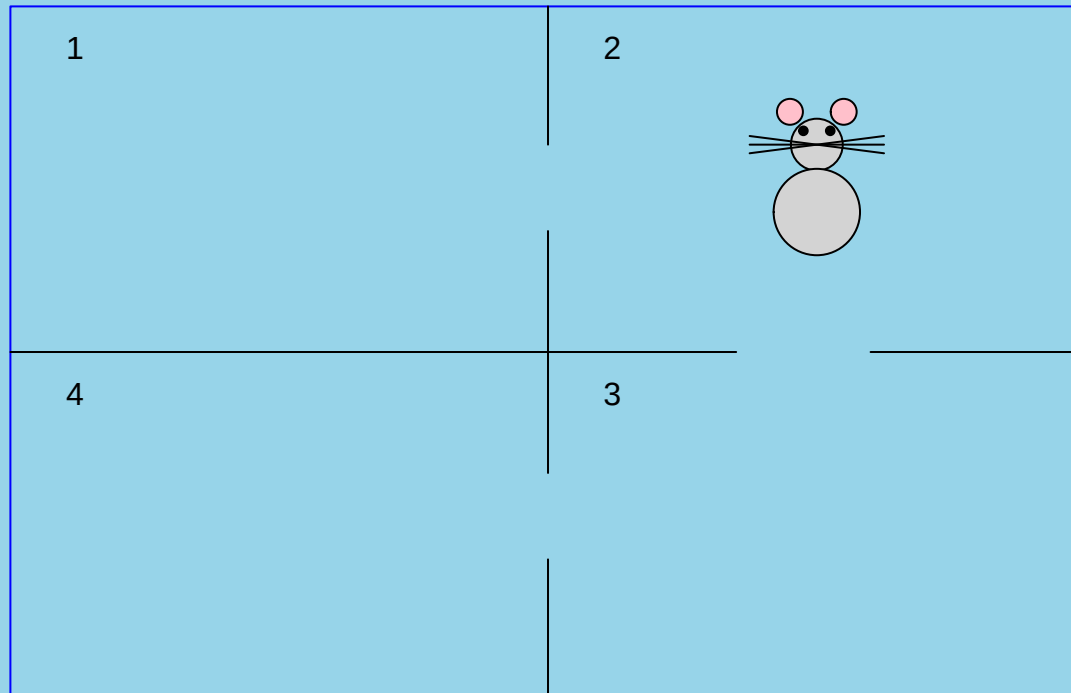
Simulating from the Mouse Movement Model

X_t = compartment at time t

$X_0 = 1$ (Mouse starts in the 1st compartment.)

Simulating from the Mouse Movement Model

At the first transition, the mouse enters the 2nd compartment: $X_1 = 2$.



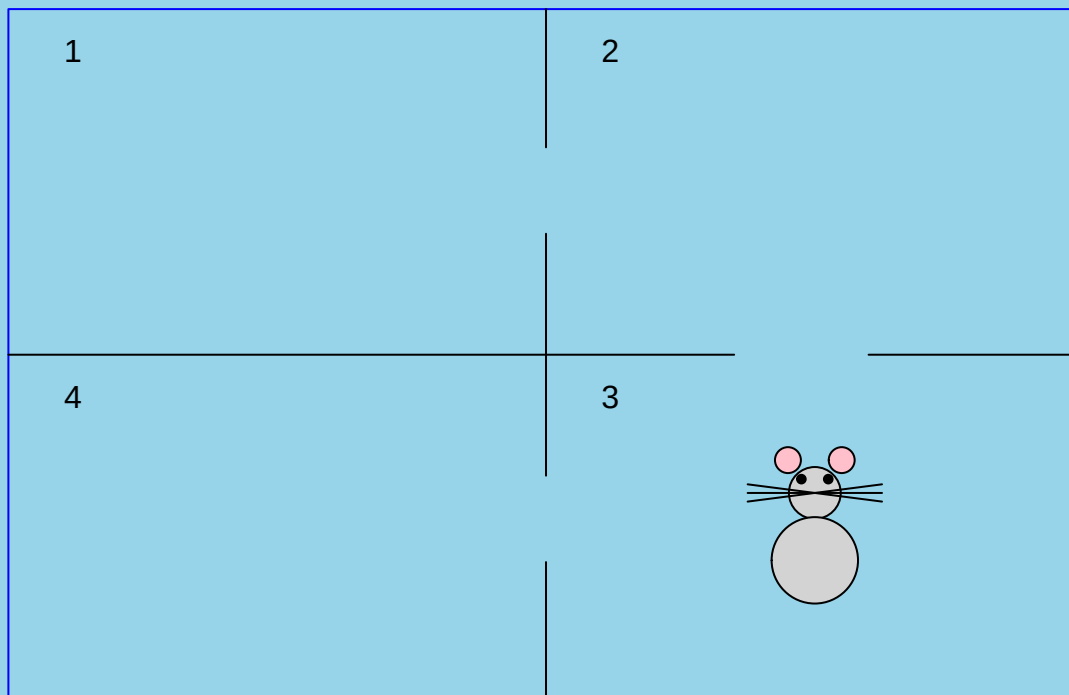
Simulating from the Mouse Movement Model

At the second transition, the mouse can enter compartment 1 with probability 0.5 or compartment 3 with probability 0.5, like a Bernoulli trial.

Generate a pseudorandom number U_2 . If $U_2 < 0.5$, the mouse enters compartment 1. Otherwise, it enters compartment 3.

Simulating from the Mouse Movement Model

$U_2 = 0.61$, so the mouse enters the 3rd compartment: $X_2 = 3$.



Simulating from the Mouse Movement Model

$U_3 = 0.34$, **so** the mouse enters the 2nd compartment: $X_3 = 2$.

$U_4 = 0.88$, **so** $X_4 = 3$.

$U_5 = 0.52$, **so** $X_5 = 4$.

The mouse must re-enter compartment 3 with probability 1: $X_6 = 3$

and so on

$\{X_1, X_2, X_3, \dots\}$ is an example of a Markov chain.

Simulating the Model in R

Using the `sample()` function to simulate data from a discrete probability distribution

$$P(X = 1) = .1, P(X = 2) = .2, P(X = 3) = .1, P(X = 4) = .4, P(X = 5) = .2$$

```
sample(1:5, size = 10, replace = TRUE,
      prob = c(.1, .2, .1, .4, .2))
```

```
## [1] 4 1 4 2 2 2 4 4 4 4
```

Note that when simulating values in a Markov chain, we are simulating from probability distributions defined by the rows of the transition matrix.

Simulating the Model in R

```
Ntransitions <- 100000 # number of mouse moves
P0 <- matrix(c(0, 1, 0, 0,
               0.5, 0, 0.5, 0,
               0, 0.5, 0, 0.5,
               0, 0, 1, 0), nrow = 4,
             byrow = TRUE) # P0 is the transition matrix
location <- numeric(Ntransitions) #initializing the chain
current.state <- 1 # initial compartment
for (t in 1:Ntransitions) {
  current.state <- sample(1:4,
                        size = 1, prob = P0[current.state, ])
  location[t] <- current.state
}
```

Simulating the Model in R

```
table(location) # this counts visits to each compartment
```

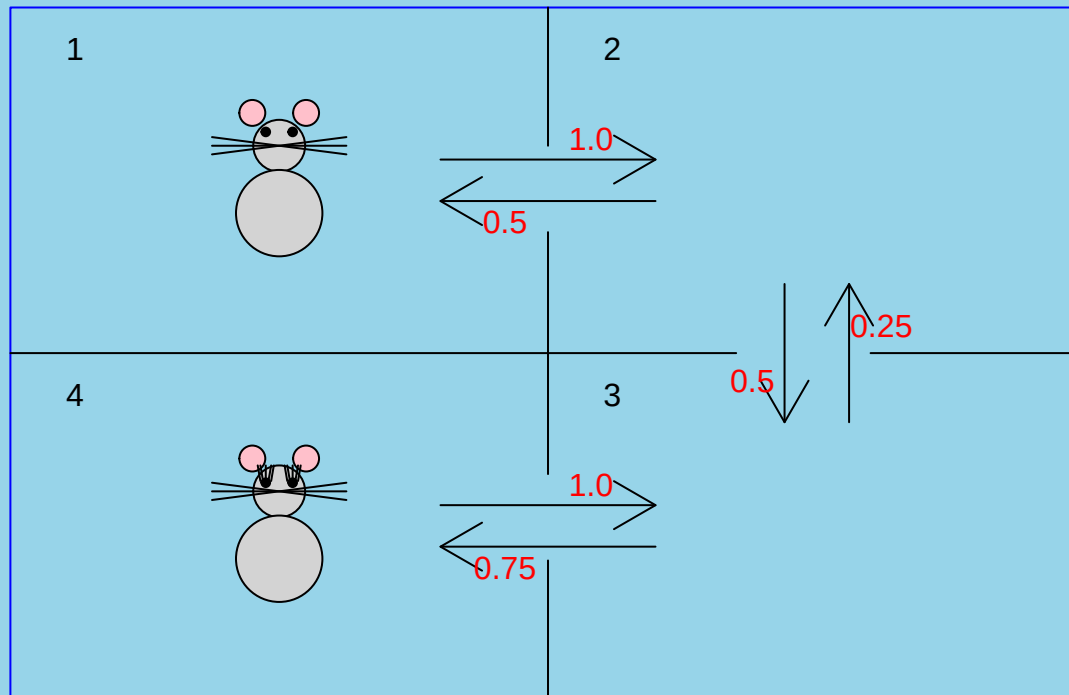
```
## location
```

```
##      1      2      3      4
```

```
## 16558 33173 33442 16827
```

A Mouse Movement Model - Including Some Structure

Transition Probabilities:



A Mouse Movement Model - Some Structure

Transition Matrix

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0.5 & 0 & 0.5 & 0 \\ 0 & 0.25 & 0 & 0.75 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Simulating the More Structured Model in R

```
Ntransitions <- 100000 # number of mouse moves
P <- matrix(c(0, 1, 0, 0,
              0.5, 0, 0.5, 0,
              0, 0.25, 0, 0.75,
              0, 0, 1, 0), nrow = 4,
            byrow = TRUE) # P is the transition matrix
location <- numeric(Ntransitions) # initializing
current.state <- 1 # initial compartment
for (t in 1:Ntransitions) {
  current.state <- sample(1:4,
                        size = 1, prob = P[current.state, ])
  location[t] <- current.state
}
```

Simulating the More Structured Model in R

```
table(location)
```

```
## location
```

```
##      1      2      3      4
```

```
## 10090 20112 39910 29888
```

```
# the odor in compartment 4 makes
```

```
# this compartment more attractive
```

Definitions and Terminology

Definition: *State Space*: the set S of all possible values that the random variable X can take.

For a Markov chain with m states, we usually denote this as
 $S = \{1, 2, \dots, m\}$.

e.g. For the mouse example, $S = \{1, 2, 3, 4\}$.

Definitions and Terminology

Definition: The sequence of random variables $X_1, X_2, X_3, \dots$, is called a **Markov chain** if

$$\begin{aligned} P(X_n = j_n | X_{n-1} = j_{n-1}, X_{n-2} = j_{n-2}, \dots) \\ = P(X_n = j_n | X_{n-1} = j_{n-1}) \end{aligned}$$

where $j_n, j_{n-1}, j_{n-2}, \dots$ **are elements of** S .

This is the discrete-time Markov property.

Definitions and Terminology

Define a matrix P with (i, j) th entry

$$p_{ij} == P(X_n = j | X_{n-1} = i)$$

p_{ij} is called the transition probability from state i to state j . P is called a transition matrix. (We are assuming that p_{ij} does not depend on n .)

Definitions and Terminology

All rows of P sum to one. That is,

$$P \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$$

A 3×3 Example

$$P_{33} = \begin{bmatrix} 0 & 0.4 & 0.6 \\ 0.5 & 0 & 0.5 \\ 0.25 & 0 & 0.75 \end{bmatrix}$$

```
P33 <- matrix(c(0, 0.4, 0.6,  
               0.5, 0, 0.5,  
               0.25, 0, 0.75), nrow = 3,  
              byrow = TRUE)
```

A 4×4 Example

$$P_{44} = \begin{bmatrix} 0.5 & 0.5 & 0 & 0 \\ 0.5 & 0.5 & 0 & 0 \\ 0 & 0 & 0.5 & 0.5 \\ 0 & 0 & 0.5 & 0.5 \end{bmatrix}$$

Another 4×4 Example

$$P_{44m} = \begin{bmatrix} 0.5 & 0.4 & 0.1 & 0 \\ 0.5 & 0.5 & 0 & 0 \\ 0 & 0 & 0.5 & 0.5 \\ 0 & 0 & 0.5 & 0.5 \end{bmatrix}$$

```
P44m <- matrix(c(0.5, 0.4, 0.1, 0,
                  0.5, 0.5, 0, 0,
                  0, 0, 0.5, 0.5, 0, 0, 0.5, 0.5), nrow = 4,
                byrow = TRUE)
```

P44m

```
##      [,1] [,2] [,3] [,4]
## [1,] 0.5 0.4 0.1 0.0
## [2,] 0.5 0.5 0.0 0.0
## [3,] 0.0 0.0 0.5 0.5
## [4,] 0.0 0.0 0.5 0.5
```

Yet Another 4×4 Example

$$P_{44m2} = \begin{bmatrix} 0.5 & 0.4 & 0.1 & 0 \\ 0.5 & 0.5 & 0 & 0 \\ 0 & 0.1 & 0.4 & 0.5 \\ 0 & 0 & 0.5 & 0.5 \end{bmatrix}$$

```
P44m2 <- matrix(c(0.5, 0.4, 0.1, 0,
                  0.5, 0.5, 0, 0,
                  0, .1, 0.4, 0.5, 0, 0, 0.5, 0.5), nrow = 4,
                  byrow = TRUE)
```

P44m2

```
##      [,1] [,2] [,3] [,4]
## [1,] 0.5 0.4 0.1 0.0
## [2,] 0.5 0.5 0.0 0.0
## [3,] 0.0 0.1 0.4 0.5
## [4,] 0.0 0.0 0.5 0.5
```

Probability Calculations

Theorem 0:

$P(X_n = j | X_0 = i)$ is the (i, j) th element of the matrix P^n .

Example. Consider the mouse (odor) maze.

```
P2 <- P%*%P # matrix multiplication of P by itself
P2

##           [,1]  [,2]  [,3]  [,4]
## [1,] 0.500 0.000 0.500 0.000
## [2,] 0.000 0.625 0.000 0.375
## [3,] 0.125 0.000 0.875 0.000
## [4,] 0.000 0.250 0.000 0.750
```

The probability of returning to compartment 1 after 2 transitions is 0.5.
The probability of reaching compartment 4 in 2 transitions, if starting in compartment 2, is 0.375.

Mouse Odor Model Example

Example. Transition probabilities at times 3 and 4:

```
P3 <- P2%*%P;  P4 <- P3%*%P
```

P3

##		[, 1]	[, 2]	[, 3]	[, 4]
##	[1,]	0.0000	0.62500	0.0000	0.37500
##	[2,]	0.3125	0.00000	0.6875	0.00000
##	[3,]	0.0000	0.34375	0.0000	0.65625
##	[4,]	0.1250	0.00000	0.8750	0.00000

P4

##		[, 1]	[, 2]	[, 3]	[, 4]
##	[1,]	0.312500	0.000000	0.687500	0.000000
##	[2,]	0.000000	0.484375	0.000000	0.515625
##	[3,]	0.171875	0.000000	0.828125	0.000000
##	[4,]	0.000000	0.343750	0.000000	0.656250

Probability Calculations

Define

$$x^{\{n\}} = [P(X_n = 1) \ P(X_n = 2) \ \dots \ P(X_n = m)]$$

This vector is called the n th state vector of the Markov chain.

It specifies the probability distribution of X_n .

The sum of the entries of $x^{\{n\}}$ must always be one. $x^{\{0\}}$ denotes the distribution of the initial state X_0 .

Probability Calculations

For the mouse example, if the mouse starts in compartment 1,
 $x^{\{0\}} = [1, 0, 0, 0]$.

If the mouse starts in a randomly selected compartment,
 $x^{\{0\}} = [.25, .25, .25, .25]$.

Theorem 1:

$$x^{\{n\}} = x^{\{0\}} P^n$$

Example - $x^{\{0\}} P_{33}$ and $x^{\{0\}} P_{33}^2$ with random starting point

```
x0 <- rep(1/3, 3) # random starting point
x1 <- x0 %*% P33   # distribution after 1 transition
x2 <- x0 %*% (P33 %*% P33) # distribution after 2 transitions
```

Probability Calculations - Example (cont'd)

$x^{\{0\}}$, $x^{\{1\}}$ and $x^{\{2\}}$:

x_0

```
## [1] 0.3333333 0.3333333 0.3333333
```

x_1

```
##          [, 1]          [, 2]          [, 3]
## [1, ] 0.25 0.1333333 0.6166667
```

x_2

```
##          [, 1] [, 2]          [, 3]
## [1, ] 0.2208333 0.1 0.6791667
```

Probability Calculations

Example. Mouse Maze distributions at times 2, 3 and 4:

```
x0 <- c(1, 0, 0, 0); x2 <- x0%*%P2; x3 <- x0%*%P3;
```

```
x4 <- x0%*%P4
```

```
x2
```

```
##      [,1] [,2] [,3] [,4]
```

```
## [1,] 0.5  0   0.5  0
```

```
x3
```

```
##      [,1] [,2] [,3] [,4]
```

```
## [1,] 0 0.625 0 0.375
```

```
x4
```

```
##      [,1] [,2] [,3] [,4]
```

```
## [1,] 0.3125 0 0.6875 0
```

Probability Distribution of X_n When n is Large

Consider a Markov chain with transition matrix P_{33} and a random initial state.

```
x0 <- rep(1/3, 3) # initial state vector
P2 <- P33%*%P33
P4 <- P2%*%P2 # 4th power of P33
P8 <- P4%*%P4 # 8th power of P33
P16 <- P8%*%P8 # 16th power of P33
x16 <- x0%*%P16 # distribution after 16 transitions
x16

##           [,1]      [,2]      [,3]
## [1,] 0.2173913 0.08695657 0.6956522

x32 <- x16%*%P16 # distribution after 32 transitions
x32

##           [,1]      [,2]      [,3]
## [1,] 0.2173913 0.08695652 0.6956522
```

The distribution of X_n no longer seems to depend on n . We have found a limiting **stationary distribution** or **equilibrium distribution** of this Markov chain.

For this Markov chain, this is the only stationary distribution.

Does Every Markov Chain Have a Unique Stationary Distribution?

No.

But Markov chains with *regular* transition matrices do.

Definition: P is said to be a regular if there exists some positive integer n such that all entries of P^n are greater than zero.

Regular Transition Matrices

Example

$P_{33} \% * \% P_{33} \% * \% P_{33}$ # 3rd power of P_{33}

##		[, 1]	[, 2]	[, 3]
##	[1,]	0.162500	0.140	0.697500
##	[2,]	0.268750	0.050	0.681250
##	[3,]	0.228125	0.075	0.696875

P_{33} is a regular transition matrix.

Regular Transition Matrices

Example

$P_{44m2} \% * \% P_{44m2} \% * \% P_{44m2}$ # 3rd power of P_{44m2}

```
##          [,1]  [,2]  [,3]  [,4]
## [1,] 0.430 0.394 0.106 0.070
## [2,] 0.475 0.430 0.070 0.025
## [3,] 0.070 0.106 0.394 0.430
## [4,] 0.025 0.070 0.430 0.475
```

P_{44m2} is a regular transition matrix.

Regular Transition Matrices

Example

$P_{44m} \circledast P_{44m} \circledast P_{44m}$ # 3rd power of P_{44m}

##		[, 1]	[, 2]	[, 3]	[, 4]
##	[1,]	0.425	0.380	0.120	0.075
##	[2,]	0.475	0.425	0.075	0.025
##	[3,]	0.000	0.000	0.500	0.500
##	[4,]	0.000	0.000	0.500	0.500

P_{44m} is not a regular transition matrix.

Regular Transition Matrices

Example: Recall

$$P_{44} = \begin{bmatrix} 0.5 & 0.5 & 0 & 0 \\ 0.5 & 0.5 & 0 & 0 \\ 0 & 0 & 0.5 & 0.5 \\ 0 & 0 & 0.5 & 0.5 \end{bmatrix}$$

$$P_{44}P_{44} = P_{44}.$$

This means that

$$P_{44}^n = P_{44}$$

for $n = 1, 2, \dots$

Since P_{44} has some 0 entries, P_{44} is not a regular transition matrix. It has many stationary distributions.

Stationary Distribution

P for the mouse odor maze is not regular.

Starting from a given initial distribution, what are the distributions at times 16 and 17?

```
x0 <- c(1, 0, 0, 0) # initial state vector
P2 <- P%*%P; P4 <- P2%*%P2; P8 <- P4%*%P4
P16 <- P8%*%P8; x16 <- x0%*%P16
x16 # probability distribution at time 16

##           [,1] [,2]           [,3] [,4]
## [1,] 0.2003129      0 0.7996871      0

x17 <- x16%*%P
x17 # probability distribution at time 17

##           [,1]           [,2] [,3]           [,4]
## [1,]      0 0.4002346      0 0.5997654
```

x_{16} and x_{17} are very different. The state vectors are not converging.

Eventually, x_{2n} converges to $[1 \ 0 \ 4 \ 0]/5$ and x_{2n+1} converges to $[0 \ 2 \ 0 \ 3]/5$.

This Markov chain is said to be periodic (with period 2).

Stationary Distribution

We will see later that $[.1.2.4.3]$ is a stationary distribution for the mouse odor example. How do the state vectors evolve through time now?

```
x0 <- c(.1, .2, .4, .3) # initial state vector
x1 <- x0%*%P; x1 # distribution at time 1

##      [,1] [,2] [,3] [,4]
## [1,]  0.1  0.2  0.4  0.3

x2 <- x1%*%P; x2 # distribution at time 2

##      [,1] [,2] [,3] [,4]
## [1,]  0.1  0.2  0.4  0.3

x3 <- x2%*%P; x3 # distribution at time 3

##      [,1] [,2] [,3] [,4]
## [1,]  0.1  0.2  0.4  0.3
```

Even though the Markov chain is periodic, if it is initialized according to the stationary distribution, the subsequent state vectors all follow the stationary distribution.

Stationary Distribution

Theorem 2: If P is a regular matrix, then there exists a vector $\mathbf{q}$ such that

$$\lim_{n \rightarrow \infty} P^n = \begin{bmatrix} \mathbf{q} \\ \vdots \\ \mathbf{q} \end{bmatrix}$$

Stationary Distribution

Example

```
library(matrixcalc) # contains function for computing matrix  
P33_20 <- matrix.power(P33, 20) # 20th power of P33  
P33_20
```

```
##           [,1]      [,2]      [,3]  
## [1,] 0.2173914 0.08695641 0.6956522  
## [2,] 0.2173912 0.08695661 0.6956522  
## [3,] 0.2173913 0.08695654 0.6956522
```

$\mathbf{q} = [0.2173, 0.08695, 0.6956]$

Stationary Distribution

Example

```
P44m2power <- matrix.power(P44m2, 150) # 150th power of P44m2
```

```
P44m2power
```

```
##      [,1] [,2] [,3] [,4]
## [1,] 0.25 0.25 0.25 0.25
## [2,] 0.25 0.25 0.25 0.25
## [3,] 0.25 0.25 0.25 0.25
## [4,] 0.25 0.25 0.25 0.25
```

$\mathbf{q} = [0.25, 0.25, 0.25, 0.25]$

Stationary Distribution Example

P16 # 16th power of P for Mouse Maze with Odor

```
##           [,1]      [,2]      [,3]      [,4]
## [1,] 0.2003129 0.0000000 0.7996871 0.0000000
## [2,] 0.0000000 0.4002346 0.0000000 0.5997654
## [3,] 0.1999218 0.0000000 0.8000782 0.0000000
## [4,] 0.0000000 0.3998436 0.0000000 0.6001564
```

matrix.power(P16, 3)

```
##           [,1] [,2] [,3] [,4]
## [1,] 0.2 0.0 0.8 0.0
## [2,] 0.0 0.4 0.0 0.6
## [3,] 0.2 0.0 0.8 0.0
## [4,] 0.0 0.4 0.0 0.6
```

Conditions of Theorem 2 are not satisfied.

Stationary Distribution Example

P0 # *Original Mouse Maze (without odor)*

```
##           [,1] [,2] [,3] [,4]
## [1,]    0.0  1.0  0.0  0.0
## [2,]    0.5  0.0  0.5  0.0
## [3,]    0.0  0.5  0.0  0.5
## [4,]    0.0  0.0  1.0  0.0
```

matrix.power(P0, 20)

```
##           [,1]      [,2]      [,3]      [,4]
## [1,] 0.333334 0.000000 0.666666 0.000000
## [2,] 0.000000 0.666667 0.000000 0.333333
## [3,] 0.333333 0.000000 0.666667 0.000000
## [4,] 0.000000 0.666666 0.000000 0.333334
```

Again, conditions of Theorem 2 are not satisfied.

Stationary Distribution

Theorem 3: If P is a regular matrix, then there exists a unique vector $\mathbf{q}$ such that

$$\lim_{n \rightarrow \infty} x^{\{n\}} = \mathbf{q}$$

for any initial state vector $x^{\{0\}}$.

This result follows immediately from Theorem 2.

It implies that, a Markov chain with a regular transition matrix converges to its unique stationary distribution. It does not matter where the Markov chain starts.

Stationary Distribution

Theorem 4: If P is a regular matrix, then the Stationary distribution vector q is the unique solution to the equation

$$q = qP$$

whose entries sum to one. The solution of the above equation is also called the *stationary* vector.

This follows from Theorems 2 and 3.

If P is not regular, it is possible for $q = qP$ to hold for a *stationary* vector q but where q is not the limiting distribution when $x_0 \neq q$.

P for the mouse maze example has a stationary distribution but this distribution is not the limiting distribution unless $x_0 = q$.

Stationary Distribution

Example

```
q <- c(.2173913, .08695652, 0.6956522)
q%*%P33 # test to see if equality holds here

##           [,1]      [,2]      [,3]
## [1,] 0.2173913 0.08695652 0.6956522
```

Mouse Odor Example

This is a non-regular example, but

```
q <- c(.1, .2, .4, .3)
q%*%P # test to see if equality holds here

##      [,1] [,2] [,3] [,4]
## [1,]  0.1  0.2  0.4  0.3
```

the stationary distribution still satisfies

$$\mathbf{q} = \mathbf{q}P.$$

Law of Large Numbers for Markov Chains

Theorem 5: If $X_1, X_2, \dots$ is a finite state Markov chain with a regular transition matrix, then for any function $f(j)$ defined on the state space, and for any initial state X_0 ,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n f(X_k) = \mathbf{q} \begin{bmatrix} f(1) \\ \vdots \\ f(m) \end{bmatrix} = \sum_{j=1}^m q_j f(j) \quad \text{with probability 1}$$

Note that

$$E[f(X)] = \sum_{j=1}^m q_j f(j)$$

when X has a distribution given by $\mathbf{q}$. (One implication of this is Markov Chain Monte Carlo simulation, i.e. MCMC).

Law of Large Numbers for Markov Chains

Example. Consider the $P33$ Markov chain. If we set $f(x) = 1$ when $x = 2$, and $f(x) = 0$, otherwise, the law of large numbers says that the average $\frac{1}{n} \sum_{k=1}^n f(X_k)$, the observed proportion of 2's, should converge to q_2 , which is .0869. The simulated X's below are closing in on this value.

```
Ntransitions <- 1000000
P33 <- matrix(c(0, 0.4, 0.6,
                0.5, 0, 0.5,
                0.25, 0, 0.75), nrow = 3,
              byrow = TRUE)
state <- numeric(Ntransitions) #initializing the Markov chain
current.state <- sample(1:3, size = 1) # initial state
for (t in 1:Ntransitions) {
  current.state <- sample(1:3,
                        size = 1, prob = P33[current.state, ])
  state[t] <- current.state
}
table(state)/Ntransitions # proportion of time spent in each state

## state
##      1      2      3
## 0.217266 0.087177 0.695557
```

Law of Large Numbers for Periodic Markov Chains

The Law of Large Numbers also holds for periodic Markov chains as long as the matrix of transitions within each periodic class is regular.

Example: $f(x) = x^2$. Check result for Mouse Odor example:

```
mean(location^2) # location contains a simulated chain

## [1] 9.27936

(1:4)^2*prob(.1, .2, .4, .3) # expected value of f(X^2)

## [1] 9.3
```

The two results match.

Classification of Markov Chain States

The following discussion leads to a simple way of determining whether a transition matrix is regular or not.

Definition. State i *leads* to state j if there exists $n \geq 1$ such that $P_{ij}^{(n)} > 0$.

e.g. Mouse maze: State 1 leads to state 2; State 2 leads to state 1, etc.

Classification of Markov Chain States

Definition. States i and j are said to *communicate* if i leads to j and j leads to i .

e.g. States 1 and 2 communicate.

Classification of Markov Chain States

Proposition. The ‘leads to’ relation is transitive. That is, if i leads to j and j leads to k , then i leads to k .

Mouse Maze e.g.: State 1 leads to State 2, State 2 leads to State 3, so State 1 leads to State 3. Similarly, State 3 leads to State 1. Therefore, States 1 and 3 communicate. Similarly, State 1 and State 4 communicate.

Classification of Markov Chain States

Definition. A *class* of states is defined as a subset of S in which any two members communicate.

Mouse Maze e.g.: All states communicate. They form a class.

Classification of Markov Chain States

Example:

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 0.5 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

States 1 and 2 communicate, but State 3 does not communicate with States 1 and 2. $\{1, 2\}$ is one class and $\{3\}$ is another class.

Classification of Markov Chain States

Definition. S is said to be *irreducible* if it is a class. That is, if all states in S communicate, S is said to be irreducible.

The mouse maze state space is irreducible, but the state space for P is not irreducible.

Classification of Markov Chain States

Definition. For any $i \in S$, the *period* of state i is defined to be the greatest common divisor of the set

$$\{n > 0 : P_{ii}^{(n)} > 0\}$$

Proposition. If i and j communicate, then the periods of i and j are the same.

Classification of Markov Chain States

Definition. If the state space of a Markov chain is irreducible, then the period of the Markov chain is defined to be the common period of each state.

Mouse Maze e.g.: Period is 2, since if the chain starts in State 1 it can never return to State 1 in an odd number of transitions, and all states communicate.

Classification of Markov Chain States

Definition. If the period of a Markov chain is 1, the Markov chain is said to be aperiodic.

P_{33} is aperiodic.

Classification of Markov Chain States

Theorem. An aperiodic irreducible Markov chain with a finite state space must have a regular transition matrix.

P_{33} is regular. P for the mouse maze example is not regular.

Calculation of Stationary Vector

The Stationary vector $\mathbf{q}$ solves

$$\mathbf{q} = \mathbf{q}P$$

which can be re-written as

$$(P^{\top} - I)\pi = 0$$

where $\pi = \mathbf{q}^{\top}$.

Calculation of Stationary Vector

Example:

```
P33 <- matrix(c(0, 0.4, 0.6,
                0.5, 0, 0.5,
                0.25, 0, 0.75), nrow = 3,
              byrow = TRUE)
A <- t(P33) - diag(rep(1,3)) #  $P^T - I$ 
solve(A, rep(0,3)) # solve  $A \pi = 0$ 
```

```
## Error in solve.default(A, rep(0, 3)): Lapack routine
dgesv: system is exactly singular: U[3,3] = 0
```

trouble! We either have too many solutions or no solutions to this problem. $P^T - I$ is singular, so there are too many solutions. We need more equations.

Calculation of Stationary Vector

Since the Stationary vector is a (discrete) probability distribution, its elements must sum to 1:

$$\pi_1 + \pi_2 + \pi_3 = 1, \quad \text{so we include this equation:}$$

```
A <- rbind(A, rep(1, 3))  
RHS <- c(rep(0, 3), 1)  
qr.solve(A, RHS)  # no longer a square system  
  
## [1] 0.21739130 0.08695652 0.69565217
```

Solution of Linear Systems via QR

Every matrix A has a QR decomposition.

Suppose A is $n \times m$ with $n > m$ (as in our case, where $n = m + 1$).

$$A = QR$$

where Q is an $n \times n$ orthogonal matrix, i.e. $Q^T Q = I$, and R is an $n \times m$ upper triangular matrix. Among other things, this means that all entries of R 's bottom $n - m$ rows are 0's.

Solution of Linear Systems via QR (`qr.solve()`)

Now, solve

$$Ax = y$$

$$\rightsquigarrow QRx = y$$

$$\rightsquigarrow Q^T QRx = Q^T y$$

$$\rightsquigarrow Rx = Q^T y$$

which can be solved for x using backward substitution, starting at row m .

If the last $n - m$ elements of $Q^T y$ are 0, then the system has a solution. If not, the system has no exact solution, but the result is the least-squares estimate.

Mouse Odor Example

Recall the mouse odor model:

```
P <- matrix(c(0, 1, 0, 0,
              0.5, 0, 0.5, 0,
              0, 0.25, 0, 0.75,
              0, 0, 1, 0), nrow = 4,
            byrow = TRUE) # P is the transition matrix
A <- t(P) - diag(rep(1, 4)) # P^T - I
A <- rbind(A, rep(1, 4)) # additional row
RHS <- c(rep(0, 4), 1)
qr.solve(A, RHS)

## [1] 0.1 0.2 0.4 0.3
```

An Industrial Example

At the beginning of each day, a batch of containers arrives at a stockyard having capacity to store 6 containers.

The batch size has the discrete probability distribution

$$\{q_0 = .4, q_1 = 0.3, q_2 = 0.2, q_3 = 0.1\}.$$

If the stockyard does not have sufficient space to store the whole batch, the batch as a whole is taken elsewhere.

Each day, as long as there are containers in the stockyard, exactly one container is removed from the stockyard.

An Industrial Example

1. Find the transition matrix for the Markov chain $\{X_1, X_2, \dots\}$, where X_t = the number of containers in the stockyard at the beginning of the t th day.
2. Find the Stationary distribution for this Markov chain.
3. Suppose a profit of \$100 is realized for each container that spends a night at the stockyard. Calculate the long-run average weekly profit.

An Industrial Example - Transition Matrix

X_t = the number of containers at the start of day t .

The state space is

$$S = \{0, 1, 2, 3, 4, 5\}.$$

Note that the stockyard could be empty at the beginning of a day, and because it can only hold 6 containers, it could never end a day with more than 5 containers, since 1 is always taken away.

An Industrial Example

Transition Matrix

$$P = \begin{bmatrix} 0.7 & 0.2 & 0.1 & 0 & 0 & 0 \\ 0.4 & 0.3 & 0.2 & 0.1 & 0 & 0 \\ 0 & 0.4 & 0.3 & 0.2 & 0.1 & 0 \\ 0 & 0 & 0.4 & 0.3 & 0.2 & 0.1 \\ 0 & 0 & 0 & 0.5 & 0.3 & 0.2 \\ 0 & 0 & 0 & 0 & 0.7 & 0.3 \end{bmatrix}$$

An Industrial Example

Equilibrium Distribution, i.e. Steady-State Distribution

```
P <- matrix(c(0.7, .2, .1, 0, 0, 0, 0.4, 0.3, 0.2, 0.1, 0, 0,
0, 0.4, 0.3, 0.2, 0.1, 0, 0, 0, 0.4, 0.3, 0.2, 0.1, 0, 0, 0,
0, 0, 0, 0, 0.7, 0.3), nrow=6, byrow = TRUE)
A <- t(P) - diag(rep(1, 6)) #  $P^T - I$ 
A <- rbind(A, rep(1, 6))
RHS <- c(rep(0, 6), 1)
options(digits=4)
pi <- qr.solve(A, RHS)
pi

## [1] 0.23273 0.17455 0.18909 0.18545 0.14909 0.06909
```

An Industrial Example - Expected Long-Run Daily Profit

$$\text{Profit} = 100 \times X$$

where X is the number of containers in the yard at the beginning of a day.

$$E[X] = \sum_{i=0}^5 i\pi_i$$

```
sum(pi*(0:5))
```

```
## [1] 2.051
```

so

$$E[100X] = 205.10$$

Simulating the Model in R

```
Ntransitions <- 100000 # number of transitions
ncontainers <- numeric(Ntransitions) #initializing
current.state <- 1 # initial stock
for (t in 1:Ntransitions) {
  current.state <- sample(1:6,
    size = 1, prob = P[current.state, ])
  ncontainers[t] <- current.state
}
pi <- table(ncontainers)/Ntransitions # this tabulates the
# distribution of the number of containers
pi

## ncontainers
##      1      2      3      4      5      6
## 0.22961 0.17394 0.18901 0.18547 0.15164 0.07033
```